

Customer Behavior Analysis Dashboard

1. Project Objective

The objective of this project is to analyze customer purchasing behavior and generate meaningful business insights using structured data. The project focuses on understanding customer demographics, revenue patterns, product performance, and segmentation. SQL is used for data cleaning and analysis, while Power BI is used to build an interactive dashboard for visualization and decision-making.

2. Dataset Overview

The dataset used in this project contains customer transaction data. It includes information related to customer demographics, purchasing behavior, and product details.

Key columns in the dataset include:

- Customer ID – Unique identifier for each customer
- Age – Age of the customer
- Gender – Male/Female classification
- Item Purchased – Product bought by the customer
- Category – Product category
- Purchase Amount USD – Amount spent on purchase
- Location – Customer's geographical location
- Subscription Status – Whether the customer is subscribed
- Shipping Type – Mode of delivery
- Review Rating – Customer rating for product
- Previous Purchases – Number of past purchases

The dataset was checked for inconsistencies and prepared for analysis.

3. Tools & Technologies Used

- **SQL (MySQL):** Used for data cleaning, aggregation, and analysis
- **Power BI:** Used for data visualization and dashboard creation
- **Excel/CSV:** Used as the data source

4. Data Cleaning (SQL)

Data cleaning was performed using SQL to ensure the dataset was consistent and reliable for analysis.

The following steps were carried out:

- Checked for duplicate records using grouping and counting techniques
- Verified that there were no redundant or repeated transactions
- Ensured proper formatting of columns

5. Data Analysis (SQL)

SQL was used to perform various analytical operations to extract insights from the dataset.

The following analyses were conducted:

- Calculated the **average age** of customers
- Analyzed **gender distribution** to understand customer base
- Computed **total revenue generated by each gender**
- Evaluated the impact of **subscription status on revenue**
- Identified **most frequently purchased products**
- Calculated **average product ratings**
- Segmented customers into:
 - New customers
 - Returning customers
 - Loyal customers
- Analyzed **spending behavior across different shipping types**
- Grouped customers into **age categories (Young, Adult, Senior)** and calculated revenue

6. Dashboard Overview (Power BI)

An interactive dashboard was created using Power BI to visualize key insights derived from the dataset.

The dashboard includes:

- KPI cards for quick summary
- Bar and column charts for comparisons
- Pie chart for segmentation
- Slicers for interactive filtering

Users can dynamically filter data based on:

- Gender
- Location
- Category

This allows better exploration of customer behavior.

7. Key Visuals Explanation

The dashboard consists of the following important visuals:

- **KPI Cards:**
Display Total Revenue, Total Customers, and Average Spend for quick insights
 - **Revenue by Gender:**
Compares revenue contribution between male and female customers
 - **Top 10 Products:**
Highlights the most frequently purchased products
 - **Customer Segmentation:**
Shows distribution of customers into New, Returning, and Loyal categories
 - **Revenue by Subscription Status:**
Compares spending behavior of subscribed vs non-subscribed customers
 - **Average Spend by Shipping Type:**
Analyzes how shipping methods influence spending
 - **Age Group Revenue:**
Shows revenue contribution across different age groups
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8. Key Insights

The following insights were derived from the analysis:

- Male customers contribute a higher portion of total revenue compared to female customers
- Non-subscribed customers generate more revenue than subscribed customers
- Loyal customers form the largest segment of the customer base

- A small number of products dominate overall purchases
- Spending patterns are relatively consistent across different shipping types
- Adult customers contribute the highest revenue among all age groups

These insights can help businesses improve targeting and decision-making.

9. Conclusion

This project demonstrates how SQL and Power BI can be combined to analyze structured data and build an interactive dashboard. The insights derived from the data can support business decisions related to customer targeting, product strategy, and revenue optimization.

10. Future Scope

The project can be further improved by:

- Adding time-based analysis if date data is available
- Incorporating predictive analytics for customer behavior
- Enhancing segmentation using additional features

PowerBI Dashboard:

